**1** **(a)** A rocket has an initial mass of $7.4 \times 10^6$ kg.

**(i)** Calculate the initial weight of the rocket.

weight = ..... [1]

**(ii)** Define, in words, the term weight.

.....

..... [1]

**(b)** Fig. 1.1 shows part of the speed-time graph for the rocket as it leaves the ground and travels into space.

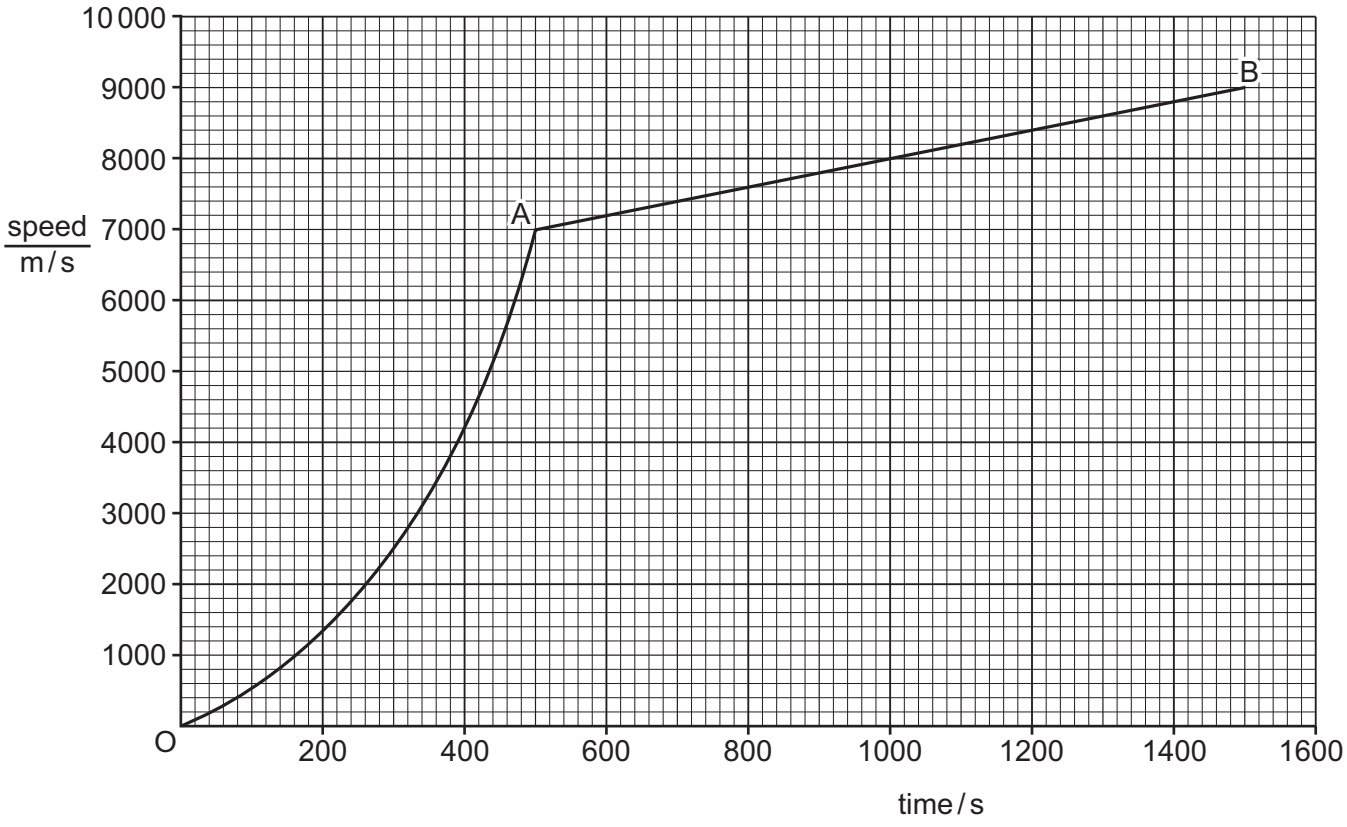


**Fig. 1.1**

**(i)** Describe the motion of the rocket:

From O to A .....

From A to B .....

[2]

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

**(ii)** Draw a tangent to the graph at time = 400 s and use this to calculate the acceleration of the rocket at this time. Show your working.

acceleration = ..... [2]

**(c)** Rockets are used to launch satellites into space. When the satellite is released, the rocket returns to the Earth.

Explain in terms of forces why the rocket reaches terminal velocity as it travels through the atmosphere back to the Earth.

.....

.....

..... [2]

[Total: 8]

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN